An elementary proof of a partition rank inequality of Mao

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Abstract. We give an elementary coefficientwise proof of Mao's inequality for the M_2-rank of partitions without repeated odd parts in the progression $3n+2$. Starting from a generating-function identity of Barman and Sachdeva, we pair consecutive terms of an alternating Lambert series. After multiplication by a partition-generating product, each pair has nonnegative coefficients. This yields an explicit lower bound by a restricted partition function and includes the progression $9n+8$, which was not covered by their partial result.

1. Introduction and statement of the result

Let λ be a nonempty partition without repeated odd parts. If $\ell(\lambda)$ is its largest part and $\nu(\lambda)$ its number of parts, its M_2-rank is

$$r_2(\lambda) = \left\lceil \frac{\ell(\lambda)}{2} \right\rceil - \nu(\lambda). \tag{1}$$

Assign the empty partition rank zero. Write $N_2(a, 6, n)$ for the number of such partitions of n with M_2-rank congruent to a modulo 6, and set

$$d(n) = N_2(0,6,n) + N_2(1,6,n) - N_2(2,6,n) - N_2(3,6,n). \tag{2}$$

The inequality considered here is $d(3n+2) > 0$ for every $n \geq 0$. Barman and Sachdeva [1] established it when $n \not\equiv 2 \pmod 3$, leaving the values $d(9m+8)$ outside the scope of their proof. This remaining progression is also the unfinished problem discussed in Chapter 4 of the April 2018 undergraduate report [3].

Define

$$S = \{k \in \mathbb{Z}_{>0} : k \not\equiv 2 \pmod 4\}, \quad P(q) = \prod_{k \in S} \frac{1}{1-q^k} = \sum_{m \geq 0} p_S(m) q^m. \tag{3}$$

Thus $p_S(m)$ counts partitions of m into odd parts and multiples of four; as usual, $p_S(0) = 1$.

Theorem 1. *For every integer $n \geq 0$,*

$$d(3n+2) \geq p_S\left(\left\lfloor \frac{n}{3} \right\rfloor\right) \geq 1. \tag{4}$$

Equivalently, for every $m \geq 0$ and $a \in \{2, 5, 8\}$,

$$d(9m+a) \geq p_S(m) \geq 1. \tag{5}$$

Jennings-Shaffer and Reihill [2, Section 7] proved the stronger result $d(n) > 0$ for every $n \geq 0$ using asymptotic formulas. Here we give an elementary proof in the progression $3n+2$, with the bound in Equation 4.

2. Reduction to coefficientwise positivity

All calculations take place in the ring $\mathbb{Q}[[q]]$ of formal power series. For $A, B \in \mathbb{Q}[[q]]$, write $A \succcurlyeq B$ if every coefficient of $A - B$ is nonnegative. Multiplication by a series with nonnegative coefficients preserves this order.

We use the standard notation

$$(a_1, ..., a_t; q)_\infty = \prod_{i=1}^{t} \prod_{j \geq 0} (1 - a_i q^j), \quad \varphi(q) = \sum_{j \in \mathbb{Z}} q^{j^2}. \tag{6}$$

In particular, $J_{3,12} = (q^3, q^9, q^{12}; q^{12})_\infty$ satisfies $\frac{1}{J_{3,12}} = P(q^3)$.

Proposition 2 (Barman–Sachdeva). *The generating function of the rank differences in Equation 2 satisfies*

$$\begin{aligned} D(q) &:= \sum_{n \geq 0} d(3n+2) q^n \\ &= \frac{P(q^3)}{q} \left[\frac{\varphi(q)^2 + \varphi(q^3)^2}{4} - \sum_{j \in \mathbb{Z}} \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}} \right]. \end{aligned} \tag{7}$$

This is the combination of Proposition 3.1 and Lemma 3.3 of [1]. It is the only previously established rank-difference identity used in the proof. The bilateral series is interpreted formally by rewriting negative-index summands with nonnegative powers of q.

For $j \geq 1$, the negative-index term has the form

$$\frac{(-1)^{-j} q^{6j^2-3j}}{1+q^{-6j}} = \frac{\left((-1)^j q^{6j^2-3j}\right) q^{6j}}{1+q^{6j}} = \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}}. \tag{8}$$

Consequently, isolating the term with index zero and pairing the remaining indices gives

$$\sum_{j \in \mathbb{Z}} \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}} = \frac{1}{2} + 2 \sum_{j \geq 1} \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}}. \tag{9}$$

Define

$$T(q) = \frac{\varphi(q)^2 + \varphi(q^3)^2 - 2}{4}, \quad L(q) = \sum_{j \geq 1} \frac{(-1)^{j+1} q^{2j^2+j}}{1+q^{2j}}. \tag{10}$$

Equation 7 therefore becomes

$$D(q) = \frac{1}{q} [P(q^3) T(q) + 2 P(q^3) L(q^3)]. \tag{11}$$

The bracketed expression has zero constant term, so division by q produces a formal power series. Each sum is locally finite: only finitely many indices can contribute to a fixed coefficient, since the minimum exponents grow quadratically.

The series $T(q)$ is manifestly nonnegative. The remaining task is to show that $P(q)L(q)$ is also nonnegative, even though the summands defining $L(q)$ alternate in sign.

3. A positivity lemma

Lemma 3. *With $P(q)$ and $L(q)$ as above, $P(q)L(q) \succcurlyeq 0$.*

Proof. For each positive odd integer r, pair the term of index r with the term of index $r+1$, and write

$$B_r(q) = P(q)\left[\frac{q^{2r^2+r}}{1+q^{2r}} - \frac{q^{2(r+1)^2+r+1}}{1+q^{2r+2}}\right]. \tag{12}$$

The difference between the two numerator exponents is $2(r+1)^2 + r + 1 - (2r^2 + r) = 4r + 3$. Taking a common denominator gives

$$B_r(q) = q^{2r^2+r}P(q)\frac{(1-q^{4r+3}) + q^{2r+2}(1-q^{4r+1})}{(1+q^{2r})(1+q^{2r+2})}. \tag{13}$$

Now factor

$$\frac{1}{(1+q^{2r})(1+q^{2r+2})} = \frac{(1-q^r)(1+q^r)(1-q^{2r+2})}{(1-q^{4r})(1-q^{4r+4})}. \tag{14}$$

For $s \in \{4r+1, 4r+3\}$, put

$$\begin{aligned} Q_{r,s}(q) &:= P(q)(1-q^r)(1-q^{2r+2})(1-q^s) \\ &= \prod_{k \in S \setminus \{r,2r+2,s\}} \frac{1}{1-q^k}. \end{aligned} \tag{15}$$

To justify the second equality, observe that r and s are odd, whereas $2r+2$ is divisible by four. All three numbers belong to S and are distinct: $r < 2r+2 < 4r+1 \leq s$. Thus the three numerator factors cancel three different denominator factors in $P(q)$. The remaining product has nonnegative coefficients, so $Q_{r,s}(q) \succcurlyeq 0$.

Substituting Equation 14 and Equation 15 into Equation 13 yields the explicit nonnegative expression

$$B_r(q) = \frac{q^{2r^2+r}(1+q^r)}{(1-q^{4r})(1-q^{4r+4})}\left[Q_{r,4r+3}(q) + q^{2r+2}Q_{r,4r+1}(q)\right]. \tag{16}$$

Every factor on the right has nonnegative coefficients. Hence $B_r(q) \succcurlyeq 0$ for every positive odd r. Summing these pairs gives

$$P(q)L(q) = \sum_{r \geq 1, r \text{ odd}} B_r(q) \succcurlyeq 0. \tag{17}$$

This regrouping is valid in $\mathbb{Q}[[q]]$, because the minimum exponent in $B_r(q)$ is $2r^2 + r$, which tends to infinity with r. □

Remark. The cancellation in Equation 15 is the role of the product $P(q)$. Positivity is proved for each paired term after multiplication by this product. No coefficientwise positivity of the alternating series $L(q)$ alone is required.

4. Proof of the main theorem

Proof of Theorem 1. Since $\varphi(q)$ has nonnegative coefficients, the definition of $T(q)$ gives

$$T(q) = q + q^2 + q^3 + q^4 + 2q^5 + \ldots \succcurlyeq q + q^2 + q^3. \tag{18}$$

Here q and q^2 arise from $\frac{\varphi(q)^2}{4}$, and q^3 arises from $\frac{\varphi(q^3)^2}{4}$. Subtracting the two constant terms in Equation 10 does not affect any positive-degree coefficient.

By Lemma 3, $P(q^3)L(q^3) \succcurlyeq 0$. It follows from Equation 11 that

$$D(q) \succcurlyeq \frac{P(q^3)}{q}(q + q^2 + q^3) = P(q^3)(1 + q + q^2). \tag{19}$$

Expanding the right-hand side using Equation 3 gives

$$P(q^3)(1 + q + q^2) = \sum_{m \geq 0} p_S(m)(q^{3m} + q^{3m+1} + q^{3m+2}). \tag{20}$$

Thus the coefficient of q^n on the right-hand side of Equation 19 is $p_S\left(\left\lfloor \frac{n}{3} \right\rfloor\right)$, proving the first inequality in Equation 4. Finally, $p_S(m) \geq 1$ for every $m \geq 0$: the partition into ones is allowed for $m > 0$, and the empty partition supplies $p_S(0) = 1$. This proves the theorem. □

Verification. The accompanying script, `verify_completion.py`, checks the generating-function identity against direct rank counts through $n = 300$ and checks its series expansion through q^{1273}. The paired product identities are checked through q^{1274}. These finite computations are independent checks of the formulas and indexing; the proof above holds for all indices without them. In the previously missing progression,

$$(d(9m + 8))_{m=0}^{9} = (1, 2, 5, 7, 9, 15, 23, 31, 40, 56). \tag{21}$$

References

[1] R. Barman and A. P. S. Sachdeva, *Proof of a limited version of Mao's partition rank inequality using a theta function identity*, Research in Number Theory **2** (2016), article 22. doi:10.1007/s40993-016-0055-5. Full text (PDF).

[2] C. Jennings-Shaffer and D. Reihill, *Asymptotic formulas related to the M_2-rank of partitions without repeated odd parts*, arXiv:1805.11319. First submitted **29 May 2018**; revised 22 February 2019. May 2018 version (PDF) · revised version (PDF) · abstract and version history. The prior proof of $d(n) > 0$ is in Section 7, inequality (7.2).

[3] *Congruences of Partitions: Rank Differences and Cubic Partition Pairs*, B.Tech. project report, Indian Institute of Technology Guwahati, April 2018, Chapter 4. Original report (PDF) · source repository.